

An Introduction to Machine Learning

Machine learning is a field of computer science that explores the study and construction of algorithms that can learn from data, based on which they can make predictions. Machine learning has evolved from pattern recognition and computational learning theory in artificial intelligence. It is used in many applications like self-driven cars, effective Web search, speech recognition, etc.

Programming is basically telling a computer what to do. Traditional programming practices assume that you understand the problem clearly and, using that understanding, will write a series of clear instructions to solve that particular problem or to do a particular task. Examples for these types of problems/tasks are many; in fact, most of the programs are written with very clear expectations of input, output and a neat algorithm for the process — for instance, sorting of numbers, removing a particular string in a text document, copying a file, etc.

However, there is a certain class of problems for which traditional programming or problem-solving methods will not be of much use. For example, let's assume that you have about 50,000 documents that have to be classified into specific categories like sports, business and entertainment — without manually going through each of them. Or consider another example of searching for a particular object in thousands of images. In the latter case, the object could be photographed in a different orientation,

or under various lighting conditions. How can you tell which images contain the object? Another very useful example is of building an online payment gateway and wanting to prevent fraudulent transactions. One way is to identify signs of potentially fraudulent transactions and triggering alerts before the transaction is complete. So how would you predict this accurately without creating unnecessary alarms?

As you can easily guess, it's not possible to write very precise algorithms for each of these problems. What we can do is to build systems that work like a human expert. A doctor can tell what disease a particular patient has by looking at the test reports. While it's not possible for him to make an accurate diagnosis 100 per cent of the time, he will be correct most times. Nobody *programmed* the doctor, but he learned those things by studying and by experience.

What we need is a system that can learn from *experience*, even when we can not *program* it to do specific tasks. What we do is, we show a large number of examples and the computer will be programmed to *learn* from this large pool of information, after which it'll be ready to predict/do the required task. This is what machine learning is all about. While the example of the doctor may be slightly different from actual machine learning, the core idea of machine learning is that machines can learn from large data sets and they improve as they *gain experience*.

